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Explainable cross-plant transfer learning for fouling prediction in data-limited pilot-scale membrane bioreactors

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ABSTRACT

Accurate membrane fouling prediction is essential for feedforward control and low-energy operation of membrane bioreactors (MBRs). However, reliable model development for data-limited plants is constrained by sparse measurements of mechanistically informative extracellular polymeric substances (EPS) and soluble microbial products (SMP). Here, we developed an explainable cross-plant transfer learning framework for transmembrane pressure (TMP) prediction in data-limited pilot-scale MBRs. Long short-term memory (LSTM) and extreme gradient boosting (XGBoost) base models were pretrained on three data-richer source MBRs treating domestic wastewater, and fine-tuned with limited data from a domestic target plant. The base models predicted source-domain TMP ($R^2 = 0.87$ and 0.86). SHapley Additive exPlanations (SHAP) identified EPS carbohydrates (EPSc), EPS proteins (EPSp), and SMP proteins (SMPp) as the dominant fouling-related drivers. Direct transfer to the target plant yielded limited accuracy ($R^2 = 0.39$ – 0.41), suggesting that the base models did not fully capture its fouling behavior. In contrast, fine-tuning improved prediction accuracy, and the fine-tuned LSTM (LSTM-FT) reached $R^2 = 0.89$ at a 40% fine-tuning ratio. SHAP and leave-one-feature-out (LOFO) analyses revealed that EPSc remained the critical transferable fouling signal, whereas EPSp became increasingly important after fine-tuning. This shift points to an EPSp-related fouling pathway linked to the plant's Fe-enriched influent. The framework was validated on a pilot-scale MBR treating industrial wastewater, where LSTM-FT achieved $R^2 \approx 0.9$, with SMPp emerging as a critical feature. Collectively, the framework preserves shared EPS-centered fouling information while adaptively recalibrating plant-specific drivers, offering a practical route for mechanism-informed fouling prediction in data-limited MBRs.

1. Introduction

Urbanization and increasing wastewater reuse requirements continue to push wastewater treatment toward intensified processes, which need to deliver stable, high-quality effluent within constrained infrastructure footprints [1]. Membrane bioreactors (MBRs), which integrate biological treatment with membrane separation, have therefore been widely adopted for domestic and industrial applications, due to their superior effluent quality and compact configuration. However, membrane fouling remains the principal obstacle to fully realizing these

advantages [2]. Under constant-flux operation, fouling progressively increases filtration resistance and is reflected by a rise in transmembrane pressure (TMP), which in turn increases aeration and pumping demand, intensifies chemical cleaning requirements, accelerates membrane aging, and raises lifecycle costs [3]. Consequently, MBRs typically consume more energy than conventional activated sludge processes (0.4 – 1.6 vs. 0.3 – 0.8 kWh m^{−3}) [4]. Accurate prediction of fouling evolution/TMP is therefore essential for enabling feedforward fouling control and low-energy MBR operation [5,6]. Nevertheless, despite substantial progress in fouling characterization and mitigation,

Học chuyển giao giải thích được giữa các nhà máy nhằm dự đoán bám bẩn trong bể phản ứng sinh học màng

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1. Giới thiệu

Trực tuyến từ ngày 19 tháng 8 năm 2026
0376-7388/© 2026 Elsevier B.V. Mọi quyền được bảo lưu, bao gồm các quyền khai thác văn bản và dữ liệu, đào tạo AI và các công nghệ tương tự.

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Đô thị hóa và các yêu cầu tái sử dụng nước thải ngày càng tăng tiếp tục thúc đẩy xử lý nước thải hướng tới các quy trình tăng cường. Các quy trình này cần cung cấp nước đầu ra ổn định, chất lượng cao trong diện tích cơ sở hạ tầng hạn chế [1]. Do đó, bể phản ứng sinh học màng (MBR) tích hợp xử lý sinh học với tách màng được ứng dụng rộng rãi cho nước thải sinh hoạt và nước thải công nghiệp. Hệ thống này mang lại chất lượng nước đầu ra vượt trội và cấu hình nhỏ gọn. Tuy nhiên, bám bẩn màng vẫn là trở ngại chính ngăn cản việc khai thác hoàn toàn các ưu điểm này [2].

proactive fouling management remains difficult in MBRs.

To support proactive management, both mechanistic and data-driven approaches have been explored to predict fouling evolution. Mechanistic and hybrid biological–physical models can enhance interpretability, but they are often challenging to parameterize and recalibrate in real plants [7]. In parallel, machine learning (ML) and deep learning (DL) models have emerged as attractive alternatives, because they can capture nonlinear relationships among operational, water-quality, and biomass-related variables [7–9]. Within this landscape, tree-based ML models (e.g., gradient-boosting families such as Extreme Gradient Boosting (XGBoost)) are attractive for tabular process data, whereas recurrent DL architectures (e.g., long short-term memory (LSTM) networks) are particularly suitable for fouling prediction, because membrane fouling is an intrinsically cumulative and path-dependent process, driven by temporal changes in sludge properties and hydrodynamic conditions [10–12].

Notably, comprehensive and rationally designed inputs for model development are key to achieving accurate AI modeling [13]. However, many existing TMP prediction models, particularly those developed at pilot-scale/full-scale MBRs, are built primarily on routinely monitored “conventional” variables, such as mixed liquor suspended solids (MLSS), mixed liquor volatile suspended solids (MLVSS), hydraulic retention time (HRT), and sludge retention time (SRT). Thus, the current approaches for tracking fouling dynamics in MBRs are often insufficient [14,15]. In contrast, mechanistically informative biochemical descriptors such as extracellular polymeric substances (EPS) and soluble microbial products (SMP) are often missing or severely under-sampled at pilot-scale/full-scale MBRs [15,16]. This is a critical limitation, because extensive evidence shows that EPS and SMP, particularly their proteinaceous and polysaccharide fractions, strongly influence fouling propensity by shaping cake formation, compressibility, and pore-blocking behavior [17], although humic and fulvic acids can also contribute to membrane fouling [18–20]. Collectively, this evidence suggests that explicitly introducing EPS and SMP as model inputs could be both scientifically justified and a promising route to improve TMP prediction and interpretability in MBR plants.

Even when these key biochemical indicators are available, pilot-/full-scale MBR datasets are often still insufficient to support the development of robust site-specific fouling prediction models. In practice, measurements of EPS and SMP fractions are usually sparse and irregular, because they rely on labor-intensive laboratory analyses rather than online sensing. Together with missing values, sensor errors, and outliers in routine operational records and key biochemical datasets, these limitations make it difficult for a **single data-limited MBR plant** to independently train a high-quality predictive model [21,22]. Most existing ML-based MBR fouling prediction models have also been developed and validated using relatively long or dense plant-specific historical records, leaving their cross-plant generalizability under data-limited conditions insufficiently established [7,8,23]. Nevertheless, different MBR plants often share comparable fouling mechanisms, including cake-layer accumulation, pore blocking, and the contribution of microbial products to membrane fouling. This suggests that **transferable models** for **fouling formation/prediction** may exist across MBR plants. From this perspective, transfer learning provides a promising strategy to address both data scarcity and data silos in MBR fouling prediction [24,25]. Specifically, a base model can be pretrained on multi-source data from different MBR plants to learn generalized relationships among operational conditions, EPS/SMP signatures, and TMP evolution. Then it could be fine-tuned using a small amount of data from the target MBR to learn the plant-specific fouling drivers.

However, cross-plant transfer learning for MBR fouling prediction is not a simple model-porting problem. First, source and target MBRs may differ substantially in influent composition, sludge physicochemical properties, EPS/SMP profiles, metal–organic interactions, and operating conditions. Moreover, even after fine-tuning, limited target plant data may still be insufficient for the model to fully capture newly dominant

fouling drivers that were weak or absent in the source plants. Furthermore, the success of transfer learning should not be evaluated by predictive accuracy alone, but also by **how** and **why** physically meaningful fouling signals are retained, discarded, or reweighted during transfer. Explainable analyses, such as SHAP and leave-one-feature-out (LOFO) analysis, are needed to determine how key EPS/SMP-related indicators function as transferable fouling drivers or plant-specific recalibrated signals. Prior applications of these methods to membrane-fouling prediction have focused mainly on model-level feature ranking [26]. However, SHAP attributions are inherently model-dependent and lack a ground truth for validation, so high predictive accuracy alone does not ensure that the highlighted features reliably reflect the true fouling behavior [27]. Accordingly, independent physicochemical characterization, including excitation–emission matrix (EEM), liquid chromatography–organic carbon detection (LC-OCD), and foulant analysis, is needed to evaluate the plausibility of these attribution shifts and support their mechanistic interpretation.

To address these gaps, this study proposes an explainable transfer learning strategy for fouling prediction in pilot-scale MBRs under data-limited conditions. Specifically, (1) base models (based on LSTM and XGBoost) were first pretrained on multi-source datasets collected from three pilot-scale MBR plants (source plants) treating domestic wastewater, covering diverse operating conditions, routine biochemical characteristics, and key components in EPS and SMP. We then (2) fine-tuned the base models using the target plant dataset to establish a fouling prediction model for the data-limited target plant (treating domestic wastewater), while quantifying the relationship between fine-tuning ratio and predictive performance and examining how LSTM and XGBoost architectures influenced transfer performance after fine-tuning. Next, (3) the transfer behavior and transferable fouling-related information were examined using explainable analyses, including SHAP and LOFO analyses, together with physicochemical characterization by EEM, LC-OCD, and foulant analysis. These analyses were used to clarify **how** key fouling-related indicators, particularly EPS and SMP, were retained or reweighted during pretraining and fine-tuning, and provided **mechanistic** support for these changes. Finally, (4) the transfer learning framework was further examined in a pilot-scale MBR treating industrial wastewater to test whether cross-plant transfer learning remains feasible under a more distinct wastewater background. Collectively, this study aims to provide a practical and explainable route for establishing fouling prediction models in data-limited MBR facilities through cross-plant transfer learning.

2. Materials and methods

2.1. Data collection

A pilot-scale MBR plant treating domestic wastewater in Singapore was selected as the target plant. The system received Fe-enriched influent, with an Fe dosage of approximately 10–30 mg/L. It had a treatment capacity of 24 m³/d and a process configuration comprising pretreatment, a five-pass step-feed bioreactor with anaerobic/oxic zonation, and a membrane tank (Fig. 1(a)). The TMP profile of the target plant during the monitoring period is shown in Fig. 2(a), the membrane module is shown in Fig. 2(b), and the detailed process and operating parameters are summarized in Text S1. The target plant dataset covered the period from September 2021 to June 2022. Owing to the twice-weekly measurement frequency of EPS and SMP, model development was based only on dates with complete matched measurements for all input variables and TMP, yielding 71 valid temporally matched records. This data-limited target plant setting reflected the practical monitoring constraints of pilot-scale MBR operation and provided a realistic scenario for evaluating the proposed cross-plant transfer learning framework.

Three pilot-scale submerged MBR plants treating domestic wastewater in Singapore were selected as source plants, yielding 332 records

2. Vật liệu và phương pháp

2.1. Thu thập dữ liệu

Một nhà máy MBR (bể phản ứng sinh học màng) quy mô pilot xử lý nước thải sinh hoạt tại Singapore được lựa chọn làm nhà máy đích (target plant). Hệ thống tiếp nhận nước đầu vào giàu Fe với liều lượng Fe khoảng 10–30 mg/L. Hệ thống có công suất xử lý 24 m³/d. Cấu hình quy trình gồm tiền xử lý, bể phản ứng sinh học cấp nước phân đoạn nằm ngấn với các vùng kỵ khí/hiếu khí và một bể màng (Fig. 1(a)). Biên dạng áp suất xuyên màng (TMP) của nhà máy đích trong thời gian theo dõi được thể hiện trong Fig. 2(a). Mô-đun màng được trình bày trong Fig. 2(b), còn quy trình và các thông số vận hành chi tiết được tóm tắt trong Text S1. Tập dữ liệu của nhà máy đích bao gồm giai đoạn từ tháng 9 năm 2021 đến tháng 6 năm 2022. Tần suất đo Chất polyme ngoại bào (EPS) và Sản phẩm vi sinh hòa tan (SMP) là hai lần mỗi tuần. Do đó, việc phát triển mô hình chỉ dựa trên các ngày có đầy đủ phép đo trùng khớp cho mọi biến đầu vào và TMP. Quá trình này mang lại 71 bản ghi hợp lệ trùng khớp theo thời gian. Bối cảnh nhà máy đích hạn chế dữ liệu này phản ánh các hạn chế giám sát thực tế trong vận hành MBR quy mô pilot. Đồng thời, bối cảnh này cung cấp một kịch bản thực tế để đánh giá khung học chuyển giao (transfer learning) liên nhà máy (cross-plant) được đề xuất.

Ba nhà máy MBR đặt ngập quy mô pilot xử lý nước thải sinh hoạt tại Singapore được chọn làm các nhà máy nguồn (source plants), mang lại 332 bản ghi tổng cộng, bao gồm 100 bản ghi từ Nhà máy 1, 132 bản ghi từ Nhà máy 2 và 100 bản ghi từ Nhà máy 3 (Fig. 1(b)). Ảnh chụp các nhà máy nguồn được cung cấp trong Fig. S1. Biên dạng TMP của các nhà máy này trong suốt thời gian theo dõi được thể hiện trong Fig. S2. Các nhà máy này có cấu hình quy trình tương đương. Cấu hình gồm xử lý sinh học thiếu khí/hiếu khí tiếp nối bởi tách màng đặt ngập, đi kèm tuần hoàn và xả bùn thông thường. Các đặc tính chi tiết của hệ thống màng và thiết lập vận hành được tóm tắt trong Table S1. Dù nhà máy đích có thành phần nước đầu vào khác biệt do bổ sung Fe ở thượng nguồn, cấu hình xử lý tổng thể vẫn tương đương với các nhà máy nguồn. Các cơ chế bám bẩn màng MBR nhìn chung tương tự nhau. Đặc điểm này ủng hộ việc áp dụng học chuyển giao (transfer learning) giữa các nhà máy nguồn và nhà máy đích.

Đối với mỗi nhà máy, tập dữ liệu bao gồm 11 biến: Thời gian lưu thủy lực (HRT), Thời gian lưu bùn (SRT), thông lượng màng (FLUX), nhu cầu oxy hóa học tổng số (TCOD), nhu cầu oxy hóa học hòa tan (SCOD), Chất rắn lơ lửng hỗn hợp bùn lỏng (MLSS), Chất rắn lơ lửng bay hơi hỗn hợp bùn lỏng (MLVSS), Protein trong EPS (EPSp), Polysaccharide trong EPS (EPSc), Protein trong SMP (SMPp) và Polysaccharide trong SMP (SMPc). Giá trị TMP được thu thập từ việc theo dõi trực tuyến thường quy và được căn chỉnh theo từng ngày lấy mẫu phục vụ phân tích tiếp theo.

2.2. Phương pháp phân tích lý hóa

2.3. Tiền xử lý dữ liệu và lựa chọn đặc trưng

Fig. 1. Khung dự đoán áp suất xuyên màng (TMP) dựa trên học chuyển giao (transfer learning) trong các hệ MBR: (a) quy trình xử lý của nhà máy đích và các thách thức dự đoán liên quan: dữ liệu hạn chế, dữ liệu khuyết thiếu và các phép đo nhiễu, (b) các đặc tính chính của ba nhà máy nguồn: kích thước tập dữ liệu lớn hơn, chất lượng dữ liệu cao hơn và cấu hình quy trình tương tự, (c) quy trình tiền huấn luyện của các mô hình Mạng bộ nhớ ngắn-dài (LSTM) và Tăng cường gradient cực đại (XGBoost) bằng dữ liệu từ ba nhà máy nguồn, (d) quy trình tinh chỉnh (fine-tuning) các mô hình cơ sở (base models) bằng tập dữ liệu nhà máy đích để thu được các mô hình học chuyển giao, (e) khả năng giải thích mô hình và phân tích thống kê bằng phân tích Loại bỏ từng đặc trưng (LOFO) và SHAP (SHapley Additive exPlanations).

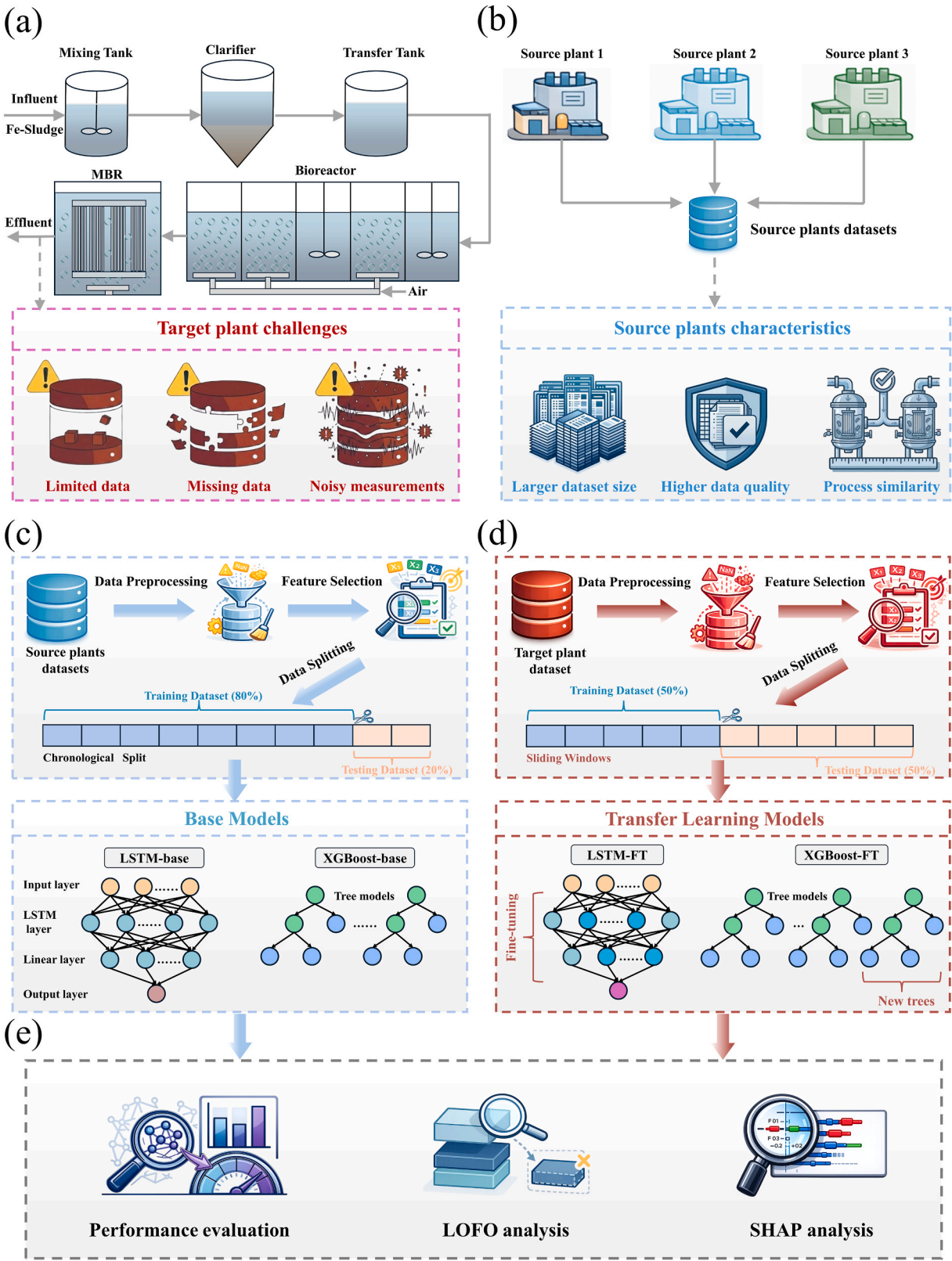


Fig. 1. Framework for transfer learning-based TMP prediction in MBRs: (a) treatment process of the target plant, and the associated prediction challenges: limited data, missing data and noisy measurements, (b) key characteristics of the three source plants: larger dataset size, higher data quality, and similar process configurations, (c) pretraining workflow of the LSTM and XGBoost models using data from the three source plants, (d) workflow for fine-tuning the base models using the target plant dataset to obtain the transfer learning models, (e) model interpretability and statistical analysis using LOFO and SHAP analyses.

Nghiên cứu thực hiện phân tích thành phần chính (PCA) đồng thời trên các biến đầu vào cùng dTMP. Phân tích này giúp đánh giá sai khác phân phối và độ chồng lấp nguồn - đích liên quan đến học chuyển giao (transfer learning). Các khối đầu vào và đầu ra được chuẩn hóa riêng biệt, đồng thời được gán trọng số tổng thể bằng nhau (Text S4).

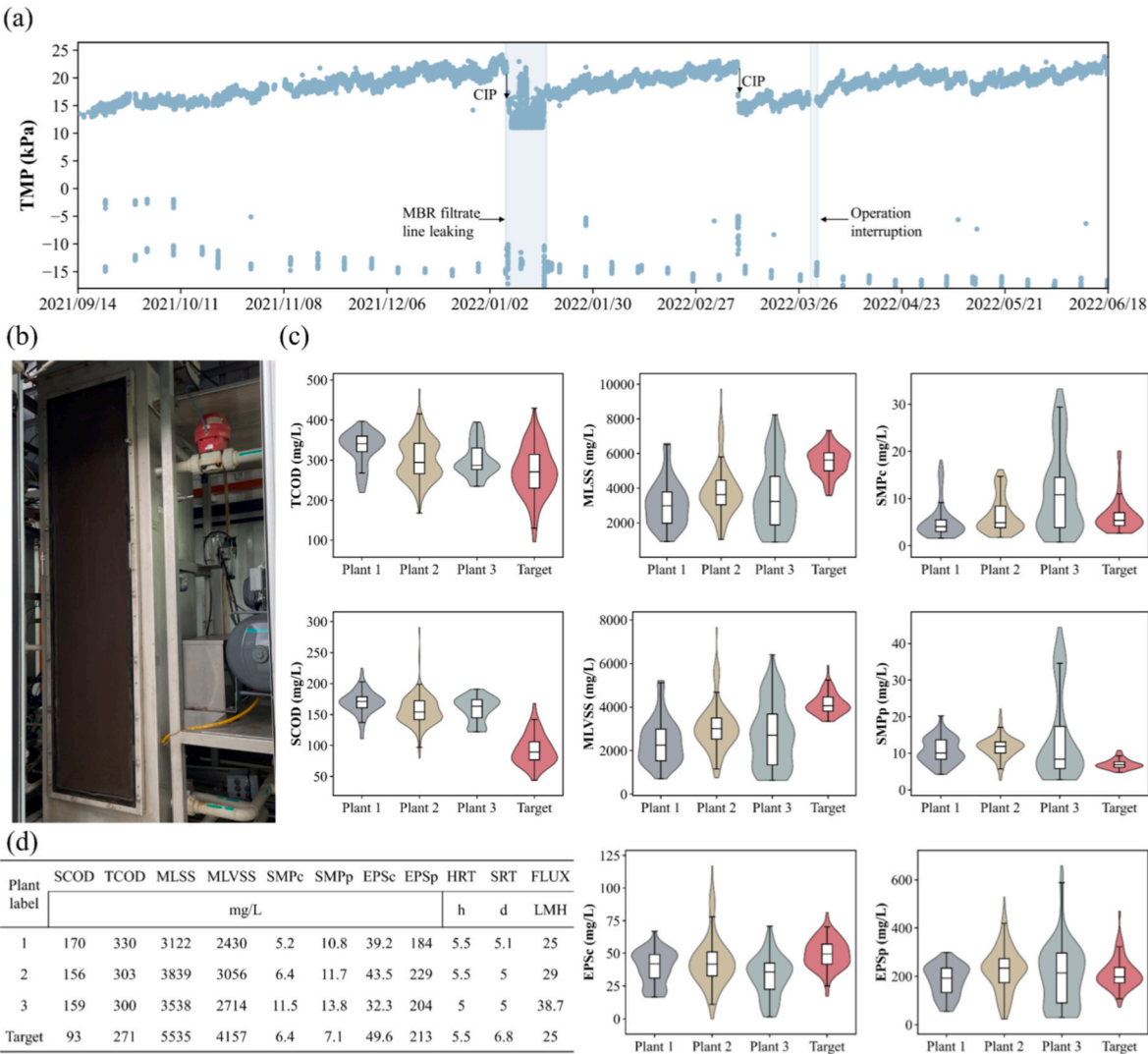


Fig. 2. Operational context of the target MBR plant and source–target dataset characteristics: (a) TMP profile of the target plant during the monitoring period, (b) photograph of the MBR module at the target plant, (c) distributions of the operational and physicochemical variables in the three source plants and the target plant, (d) mean values of these variables for each plant.

in total, including 100 records from Plant 1, 132 records from Plant 2, and 100 records from Plant 3 (Fig. 1(b)). Photographs of the source plants are provided in Fig. S1, and their TMP profiles during the monitoring periods are shown in Fig. S2. These plants had comparable process configurations, consisting of anoxic/oxic biological treatment followed by immersed membrane separation, with conventional sludge recirculation and wasting. Detailed membrane-system characteristics and operating settings are summarized in Table S1. Although the target plant differed in influent composition due to upstream Fe addition, it remained comparable to the source plants in overall treatment configuration, with broadly similar MBR fouling mechanisms, supporting the application of transfer learning between source and target plants.

For each plant, the dataset included 11 variables: HRT, SRT, membrane flux (FLUX), total and soluble chemical oxygen demand (TCOD and SCOD), MLSS, MLVSS, EPSp, EPSc, SMPp, and SMPc. TMP was obtained from routine online monitoring and aligned with each sampling date for subsequent analysis.

2.2. Physicochemical analytical methods

HRT, SRT, and FLUX were recorded as routine operational parameters. TCOD and SCOD were determined by the APHA 5220D closed

reflux colorimetric method, and MLSS and MLVSS were measured according to Standard Methods [28]. SMP was defined as the dissolved fraction of the supernatant after 0.45 μ m membrane filtration and EPS was extracted from the residual biomass using the heat method [29]. Proteins and polysaccharides in SMP and EPS, including SMPp, SMPc, EPSp, and EPSc, were quantified using the Modified Lowry Protein Assay and Dubois Method, respectively [30]. Liquid chromatography–organic carbon detection (LC-OCD) was employed to characterize the dissolved organic matter, including biopolymers with molecular weight (MW) greater than 20 kDa and low-molecular-weight (LMW) compounds with MW less than 1000 Da, such as humics, building blocks, LMW acids, and LMW neutrals. The particle size distribution (PSD) of mixed liquor was analyzed using a laser diffraction particle analyzer. Capillary suction time (CST) was measured using a CST apparatus [31]. Total organic carbon (TOC) was measured using a TOC analyzer (TOC-VCSH, Shimadzu, Japan). Excitation–emission matrix (EEM) spectra were measured using a fluorescence spectrophotometer [32]. Total iron was measured using inductively coupled plasma optical emission spectroscopy (ICP-OES).

2.6. Xây dựng các mô hình học chuyển giao (transfer learning)

2.4. Xây dựng các mô hình cơ sở (base models) dựa trên các nhà máy nguồn (source plants)

2.5. Xây dựng mô hình chuẩn đối sánh (baseline model) cho nhà máy đích (target plant)

Trước khi thiết lập các mô hình học chuyển giao (transfer learning), một mô hình chuẩn đối sánh (baseline model) được phát triển chỉ từ tập dữ liệu của nhà máy đích (target plant). Hiệu suất của mô hình này được so sánh với các mô hình học chuyển giao. Để bảo đảm tính so sánh, tập dữ liệu nhà máy đích (target plant) được phân chia theo trình tự thời gian. 50% dữ liệu đầu tiên được dùng để huấn luyện mô hình, và 50% dữ liệu còn lại được giữ lại để kiểm tra. Với cách phân chia này, lượng dữ liệu nhà máy đích (target plant) khả dụng cho việc huấn luyện mô hình rất hạn chế. Với tập huấn luyện nhỏ như vậy, mô hình chuẩn đối sánh Mạng bộ nhớ ngắn-dài (LSTM) huấn luyện riêng cho nhà máy đích dễ bị khớp không ổn định và quá khớp (overfitting). Hạn chế này làm cho kết quả so sánh giảm độ tin cậy [39]. Do đó, một mô hình XGBoost độc lập được chọn làm mô hình chuẩn đối sánh. Mô hình này được huấn luyện trên phần dữ liệu huấn luyện của nhà máy đích (target plant) với cùng tập đặc trưng như các mô hình học chuyển giao. Bộ chuẩn hóa cho biến mục tiêu cũng được khớp độc lập bằng chính phần dữ liệu huấn luyện này. Các thiết lập chi tiết được liệt kê trong Table S4.

2.3. Data preprocessing and feature selection

To ensure reliable model development, the raw datasets were first preprocessed by outlier treatment and missing-value imputation (**Text S2**). To place variables on comparable scales and prevent variables with larger numerical ranges from dominating model training, z-score standardization was applied during preprocessing (**Text S3**). The scaler was fitted using the training data only, and model outputs were inverse-transformed to the original scale after training.

The final prediction target of this study was next-day TMP, whereas dTMP was used only as an intermediate training target to alleviate non-stationarity and improve output comparability across plants [33,34]. Compared with raw TMP values, dTMP reduced inter-plant differences in TMP level and produced distributions more centered around zero (**Fig. S3**). Predicted dTMP values were then added to the current-day TMP to reconstruct next-day TMP for model evaluation and presentation.

To assess source–target distribution discrepancy and overlap relevant to transfer learning, joint principal component analysis (PCA) was performed using the input variables together with dTMP. The input and output blocks were standardized separately and assigned equal overall weights (**Text S4**).

The sample-size-to-feature ratio (SFR) is an important indicator of whether model complexity is appropriate for a given dataset size, and in small-sample modeling an SFR greater than 10 is generally considered desirable [35]. Therefore, Pearson correlation analysis was used as a coarse pre-screening step (**Text S5**). Because the final prediction target remained next-day TMP, this initial screening was performed against TMP rather than dTMP. It considered both feature–target association and inter-feature collinearity, but served only as a pragmatic dimensionality-control procedure rather than a definitive assessment of nonlinear predictive relevance.

2.4. Construction of base models based on source plants

Given the time-series nature of the data, the source datasets were split within each plant according to chronological order rather than randomly, thereby reducing the risk of information leakage between temporally adjacent samples. Considering the limited sample sizes and short input window, LSTM was selected as a compact recurrent architecture for sequence learning [36]. Transformer- and TCN-based models may offer advantages when larger MBR datasets or longer temporal contexts are available, and systematic benchmarking under such conditions remains a direction for future work [37,38]. Accordingly, two base models were developed: LSTM-base for recurrent sequence learning and XGBoost-base for tree-based learning (**Fig. 1(c)**). LSTM-base used standardized feature sequences with a sequence length of 3 time steps, whereas XGBoost-base used the same three-step historical information encoded as lagged features. This design ensured that the two base models used the same amount of historical information, allowing a fair comparison of their predictive performance. Additional implementation details are provided in **Text S6**.

Hyperparameters for both base models were optimized using the training split of the source plant datasets. Because the data were time-ordered, five-fold time-series cross-validation was combined with Optuna-based hyperparameter search to prevent leakage from future samples and improve the robustness of model selection. After optimization, both base models were trained on the same training split, and the trained models were saved to initialize the transfer learning models. The hyperparameter search spaces, final settings, and training details are provided in **Table S2–S3** and **Text S7**.

2.5. Construction of the baseline model for the target plant

Before establishing transfer learning models, a baseline model was developed using only the target plant dataset, and its performance was

compared with that of the transfer learning models. To ensure comparability, the target plant dataset was partitioned in chronological order, with the first 50% used for model training and the final 50% reserved for testing. Under this split, only a limited amount of target plant data was available for model training. With such a small training set, a standalone target-trained LSTM baseline would be more prone to unstable fitting and overfitting, making the resulting comparison less reliable [39]. An independent XGBoost model was therefore adopted as the baseline. It was trained on the training portion of the target plant dataset using the same feature set as the transfer learning models, and the scaler for the target variable was fitted independently using the same training portion. Detailed settings are listed in **Table S4**.

2.6. Construction of transfer learning models

For transfer learning, the target plant dataset was partitioned chronologically using the same 50:50 split: the first 50% was used as the fine-tuning pool, whereas the final 50% was kept unseen during fine-tuning and reserved as an independent testing set (**Fig. 1(d)**), thereby reducing the risk of temporal information leakage. The fine-tuning ratio (FT) was defined as the proportion of the total target plant dataset used for fine-tuning, and was examined at 0, 10%, 20%, 30%, 40%, and 50%. The exact numbers of target plant records in the fine-tuning pool, testing set, and each FT setting are summarized in **Table S5**. The resulting transfer models were denoted as LSTM-FT and XGBoost-FT. At FT = 0, no target plant samples were used for fine-tuning, and the two models therefore corresponded to direct transfer without target plant adaptation.

For each nonzero FT setting, multiple fine-tuning segments of the corresponding length were sampled from the fine-tuning pool to reduce dependence on any single local history (**Text S8**). Each segment-level experiment was repeated with five random seeds, and performance was reported as the mean $\pm$ standard deviation across all fine-tuning segments and repeated runs. LSTM-FT and XGBoost-FT were fine-tuned using different update strategies, with implementation details and hyperparameter settings provided in **Table S6–S7** and **Text S9**.

All experiments were implemented in Python 3.10.18 using relevant third-party libraries. The deep-learning models were built and trained in PyTorch 2.7.0. The computing environment included an Intel Core i7-14700KF processor (28 logical cores), 32 GB RAM, and an NVIDIA GeForce RTX 5060 Ti GPU.

2.7. Evaluation of model performance

Model performance was evaluated using R^2 , RMSE, and MAE (details in **Text S10**). Model interpretability was examined using LOFO and SHAP analyses (**Fig. 1(e)**), as detailed in **Text S11** and **Text S12**, respectively.

2.8. Model adaptability under cross-scenario conditions

Additional validation was conducted using data from a pilot-scale MBR treating mixed petrochemical and pharmaceutical industrial wastewater in Singapore. Detailed process configuration and operating parameters are provided in **Table S8**. This validation was performed using LSTM-FT under two fine-tuning conditions, FT = 0 and FT = 40%. To maintain consistency with the main analysis, the pretrained model, chronological fine-tuning/testing logic, data preprocessing procedure, and model settings were kept the same. Accordingly, this test was intended to assess whether the established workflow could remain effective under industrial wastewater conditions, without additional model optimization.

Tất cả các thử nghiệm được triển khai trong Python 3.10.18 cùng các thư viện bên thứ ba liên quan. Các mô hình học sâu (DL) được xây dựng và huấn luyện trong PyTorch 2.7.0. Môi trường tính toán gồm một bộ xử lý Intel Core i7-14700KF (28 lõi logic), 32 GB RAM và một GPU NVIDIA GeForce RTX 5060 Ti.

2.7. Đánh giá hiệu suất mô hình

Hiệu suất mô hình được đánh giá bằng R^2 , RMSE và MAE (chi tiết trong Text S10). Khả năng giải thích của mô hình được khảo sát bằng phân tích Loại bỏ từng đặc trưng (LOFO) và SHAP (SHapley Additive exPlanations) (**Fig. 1(e)**). Chi tiết được trình bày tương ứng trong Text S11 và Text S12.

2.8. Khả năng thích ứng của mô hình trong các điều kiện liên kịch bản

3. Kết quả và thảo luận

3.1. Phân bố nguồn–đích và phân tích thành phần chính (PCA)

Fig. 3. Phân bố dữ liệu nguồn–đích, hiện tượng cộng tuyến đặc trưng và hiệu suất dự đoán của các mô hình cơ sở (base models): (a) Phân tích thành phần chính (PCA) đồng thời của các tập dữ liệu nhà máy nguồn (source plant) và nhà máy đích (target plant), (b) bản đồ nhiệt tương quan Pearson giữa đặc trưng với đặc trưng ($X-X$), cùng các biểu đồ phân tán dự đoán cho (c) XGBoost-base và (d) LSTM-base.

Sau đó, phân tích tương quan Pearson giữa đặc trưng với đặc trưng ($X-X$) được thực hiện để kiểm soát thêm hiện tượng cộng tuyến giữa các đặc trưng (**Fig. 3(b)**). Thời gian lưu thủy lực (HRT) tương quan âm mạnh với thông lượng màng (FLUX) ($rxj,xk = -0.86$), có thể phản ánh sự liên kết vận hành giữa chúng. Cụ thể, HRT tỷ lệ nghịch với lưu lượng xử lý, còn FLUX biểu thị lưu lượng dòng thấm trên một đơn vị diện tích màng [8]. Chất rắn lơ lửng hỗn hợp bùn lỏng (MLSS) và chất rắn lơ lửng bay hơi hỗn hợp bùn lỏng (MLVSS) thể hiện tương quan dương rất mạnh ($rxj,xk = 0.96$). Điều này phù hợp với việc MLVSS là thành phần sinh khối bay hơi của MLSS [40]. Trong hai cặp đặc trưng cộng tuyến này, FLUX và MLVSS có giá trị $|rxj,y|$ với áp suất xuyên màng (TMP) cao hơn nên được giữ lại. Theo đó, các biến FLUX, thời gian lưu bùn (SRT), MLVSS, Polysaccharide trong EPS (EPS_c), Protein trong EPS (EPS_p) và Protein trong SMP (SMP_p) được giữ lại. Tập đầu vào rút gọn này phục vụ cho việc phát triển mô hình cơ sở (base model) tiếp theo.

3.2.2. Đánh giá hiệu suất của các mô hình cơ sở (base model)

3.2.3. Giải thích SHAP (SHapley Additive exPlanations) cho các mô hình cơ sở (base models) và đặc tính hóa lý

LSTM-base đạt hiệu suất dự đoán miền nguồn tốt và cho thấy nhiều triển vọng trong học chuyển giao (transfer learning) (thảo luận trong Mục 3.3). Vì vậy, các đặc tính hóa lý của sinh khối tại các nhà máy nguồn (source plant) được kiểm tra thêm. Phân tích này nhằm xác định xem mẫu quy gần đã học có phù hợp với cơ chế bám bẩn màng thực tế hay không. Chất polyme ngoại bào (EPS) và Sản phẩm vi sinh hóa tan (SMP) được chú trọng, vì chúng là các đặc trưng quan trọng trong phân tích SHAP. Cụ thể, EPS liên kết chặt chẽ hơn với các bông bùn và hiện tượng bám bẩn liên quan đến màng [45]. Trong khi đó, SMP chủ yếu đại diện cho các chất hữu cơ hòa tan [46].

3. Results and discussion

3.1. Source–target distribution and PCA analysis

The distributions of the measured variables across the source and target plants are presented in Fig. 2(c), and the corresponding average values are summarized in Fig. 2(d). At the level of core operating variables, the target plant remained within or close to the ranges of the source plants (HRT of 5.5 h versus 5.0–5.5 h, SRT of 6.8 d versus 5.0–7.0 d, and FLUX of 25 LMH versus 25–44 LMH). Across the source plants, the measured variables covered relatively broad ranges of biomass-related and fouling-related properties (e.g., MLSS of 900–9730 mg/L, MLVSS of 639–7671 mg/L, EPSp of 25.0–658.8 mg/L, EPSc of 0.0–117.1 mg/L, SMPc of 0.8–33.3 mg/L, and SMPp of 2.7–44.5 mg/L). By contrast, the target plant showed comparatively narrower distributions for the biomass-related and fouling-related variables (e.g., MLSS of 3580–7340 mg/L, MLVSS of 3340–5920 mg/L,

EPSp of 73.5–469.9 mg/L, EPSc of 17.1–81.3 mg/L, SMPc of 2.6 to 20.1 mg/L, and SMPp of 4.8–10.8 mg/L).

These distributional relationships were further examined by PCA (Fig. 3(a)). The first two principal components explained 74.0% of the total variance, indicating that the major variability in the dataset was captured in a two-dimensional projection. The source plant datasets showed substantial overlap, indicating broadly similar fouling-related variable patterns across the source plants. By contrast, the target plant dataset formed a more compact cluster with partial separation from the source plants, indicating a measurable distribution shift. Importantly, a clear overlap region remained between the target and source datasets. This overlap provided a reasonable basis for developing base models using the source plant datasets and supported the feasibility of subsequent transfer learning.

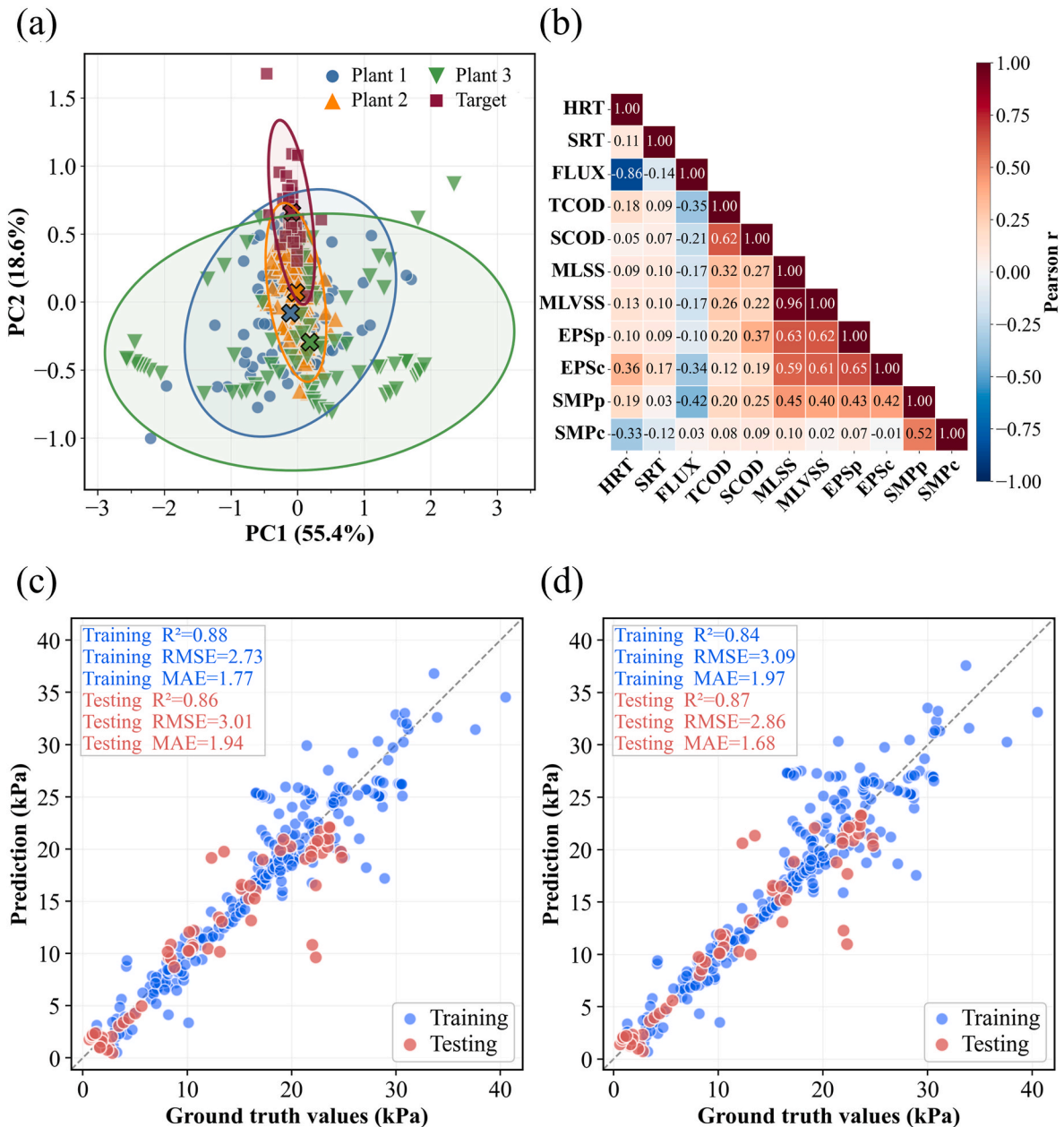


Fig. 3. Source–target data distribution, feature collinearity, and predictive performance of the base models: (a) joint PCA of the source and target plant datasets, (b) feature–feature (X–X) Pearson correlation heatmap, and prediction scatter plots for (c) XGBoost-base and (d) LSTM-base.

3.3.1. Ảnh hưởng của tỷ lệ tinh chỉnh (FT) đến hiệu suất học chuyển giao (transfer learning)

Nghiên cứu kiểm tra ảnh hưởng của tỷ lệ tinh chỉnh (FT) đến hiệu suất học chuyển giao (transfer learning). Phân tích này nhằm xác định lượng dữ liệu nhà máy đích (target plant) cần thiết để thích ứng hiệu quả. Mức FT = 0 đại diện cho chuyển giao trực tiếp không thích ứng tại nhà máy đích (target plant). Ở điều kiện này, các mô hình LSTM và XGBoost tiền huấn luyện đều cho giá trị R2 thấp, lần lượt là 0.41 và 0.39. Kết quả này phản ánh khả năng dự đoán hạn chế trong thiết lập chuyển giao zero-shot. Kết quả Phân tích thành phần chính (PCA) trước đó đã chỉ ra sự dịch chuyển phân bố đo lường được giữa nguồn và đích. Cùng với kết quả này, phát hiện trên cho thấy chỉ chuyển giao trực tiếp là chưa đủ. Mô hình không thể dự đoán áp suất xuyên màng (TMP) đáng tin cậy tại nhà máy đích (target plant) [53].

3.3. Hiệu suất và giải thích học chuyển giao (transfer learning)

3.2. Construction and evaluation of base models

3.2.1. Correlation-based feature selection

Feature–target (X–y) Pearson correlation analysis was further conducted, and the ranking results are shown in Fig. S4. Overall, FLUX, EPSc, HRT, and SMPp showed relatively strong correlations with TMP, with $|r_{x,y}|$ values of 0.471, 0.462, 0.440, and 0.347, respectively. In contrast, SCOD, SMPc, and TCOD showed weak correlations with TMP, with $|r_{x,y}|$ values all below 0.1, and were therefore excluded from the reduced input set used for subsequent modeling. The remaining variables were kept for the subsequent collinearity screening step.

Feature–feature (X–X) Pearson correlation analysis was then performed to further control inter-feature collinearity (Fig. 3(b)). HRT was strongly negatively correlated with FLUX ($r_{x_i,x_k} = -0.86$), likely reflecting their operational coupling: HRT is inversely related to treatment flow rate, whereas FLUX represents the permeate throughput per unit membrane area [8]. MLSS and MLVSS showed a very strong positive correlation ($r_{x_i,x_k} = 0.96$), consistent with MLVSS representing the volatile biomass fraction of MLSS [40]. Within these two collinear feature pairs, FLUX and MLVSS had higher $|r_{x,y}|$ values with TMP and were therefore retained. Accordingly, FLUX, SRT, MLVSS, EPSc, EPSp, and SMPp were retained as a reduced input set for subsequent base model development.

3.2.2. Performance evaluation of base models

The prediction scatter plots for both base models show that most training and testing points were distributed close to the line of equality, indicating good agreement between the predicted and observed TMP values in the source plants. XGBoost-base achieved an R^2 of 0.86, an RMSE of 3.01 kPa, and an MAE of 1.94 kPa on the test set (Fig. 3(c)). LSTM-base showed slightly better performance, with an R^2 of 0.87, an RMSE of 2.86 kPa, and an MAE of 1.68 kPa (Fig. 3(d)). Larger prediction deviations were observed at higher TMP values in both models, suggesting that TMP prediction became more difficult under more severe fouling conditions, possibly due to intensified fouling-layer compression and nonlinear fouling resistance at elevated TMP levels [41]. Overall, both models captured the main TMP variations in the source datasets, with LSTM-base showing a small but consistent advantage over XGBoost-base.

3.2.3. SHAP interpretation of base models and physicochemical characterization

SHAP attribution patterns were further analyzed for both base models. ChatGPT-5.4 was used only as an auxiliary tool to support the summarization of these attribution patterns, while the final interpretation was performed by the authors based on the SHAP plots and corresponding feature-attribution results (Fig. 6(d)). For LSTM-base, the SHAP summary plots showed that EPSc, MLVSS, EPSp, and SMPp made the largest contributions (Fig. 6(a); Fig. S5), whereas FLUX and SRT were comparatively less influential. XGBoost-base showed a much more concentrated attribution pattern, with FLUX dominating the SHAP attribution and contributing more than the other input variables (Fig. S6(a); Fig. S6(b)). These results indicate that the two base models achieved similar predictive performance but relied on different predictive structures during pretraining on the source plants. The broader dependence of LSTM-base on EPSc, MLVSS, EPSp, and SMPp may partly reflect the greater suitability of sequence-learning architectures for variables with stronger temporal continuity and gradual variation [42]. In contrast, the FLUX-dominated attribution pattern in XGBoost-base was consistent with the X–y correlation analysis and may partly reflect the tendency of tree-based models to rely heavily on a dominant predictive signal [43, 44]. These results were consistent with the subsequent transfer learning results, which showed that XGBoost-FT achieved only limited improvement compared with LSTM-FT (discussed in Section 3.3).

Given that LSTM-base achieved strong source-domain predictive

performance and showed much more promise in transfer learning (discussed in Section 3.3), the physicochemical characteristics of biomass in the source plants were further examined to determine whether its learned attribution pattern was consistent with real fouling mechanisms. EPS and SMP were emphasized, because they were identified as important features in the SHAP analysis. Specifically, EPS is more closely associated with sludge flocs and membrane-associated fouling [45], whereas SMP mainly represents soluble organics [46].

Concretely, the three source plants showed a consistent EPS-related signature, including a shared aromatic protein-like peak in the EPS EEM spectra ($E_x/E_m \approx 220\text{--}225/325\text{--}355$ nm; Fig. 4(a)–(c)), and sludge particle-size distributions dominated by particles of 20–40 μm (Fig. 4(d)). These features generally suggested a readily depositable sludge structure that could promote cake-layer formation in MBRs [47]. Moreover, the protein fraction represented by EPSp likely comprises aromatic protein-like components, which could be associated with sludge floc aggregation and the initial adhesion of foulants to the membrane surface [48]. By comparison, EPSc may be more closely related to the gel-forming and resistance-enhancing effects of the polysaccharide matrix, thereby further increasing the hydraulic resistance of the fouling layer [49]. The relatively high attribution of MLVSS (revealed by SHAP) was also physically reasonable, as MLVSS represents the biomass fraction that carries EPS and contributes to sludge deposition on the membrane surface [3].

Additionally, SMP from the three source plants exhibited similar EEM spectral features, with a shared aromatic protein-like fluorescence peak ($E_x/E_m \approx 220\text{--}230/330\text{--}350$ nm; Fig. S7(a)–(c)). These aromatic protein-like SMP components may contribute to fouling development through membrane-surface adsorption or retention within the developing fouling layer, particularly during the early stage of membrane fouling [50,51]. This interpretation was further supported by the LC-OCD results (Fig. 4(e) and (f); Table S9), which showed that SMP in the three source plants was dominated by hydrophilic DOC, with humics and biopolymers as the major fractions. For each LC-OCD fraction, rejection was calculated as the percentage decrease from its concentration in SMP to that in the corresponding effluent, relative to its concentration in SMP. On this basis, biopolymers showed the highest rejection across the three source plants, ranging from 90.0 to 95.3%. These results suggest that SMPp may mainly reflect the aromatic protein-like component within the biopolymer pool that was preferentially retained by the membrane and could therefore participate directly in organic membrane fouling [52].

Overall, the shared physicochemical characteristics of source plants provide clear mechanism-informed support for the EPS- and SMP-related attribution pattern learned by LSTM-base. This indicates that the learned structure was not merely statistical but grounded in real fouling-related properties. In turn, this supports the use of LSTM-base as a reliable pretrained model for subsequent transfer learning.

3.3. Transfer learning performance and interpretation

3.3.1. Effect of fine-tuning ratio on transfer learning performance

The effect of fine-tuning ratio (FT) on transfer learning performance was examined to determine the amount of target-plant data required for effective adaptation. At $FT = 0$, representing direct transfer without target plant adaptation, the pretrained LSTM and XGBoost models showed similarly low R^2 values of 0.41 and 0.39, respectively, indicating limited predictive power under zero-shot transfer. Together with the PCA results showing a measurable source–target distribution shift, this finding suggests that direct transfer alone was insufficient for reliable TMP prediction in the target plant [53].

For LSTM-FT, the largest gain occurred between $FT = 0$ and $FT = 10\%$, with R^2 increasing from 0.41 to 0.81 (Fig. 5(a)). As FT further increased, the performance improved more gradually, reaching R^2 , RMSE, and MAE values of 0.89, 0.50 kPa, and 0.33 kPa, respectively, at $FT = 40\%$ (Fig. 5(b) and (c)). At $FT = 50\%$, the improvement relative to

Để so sánh hiệu suất dự đoán trên nhà máy đích (target plant), mô hình chuẩn đối sánh (baseline model) cùng với LSTM-FT và XGBoost-FT đã được đánh giá (Fig. 5(d)). Mô hình chuẩn đối sánh này được huấn luyện riêng dựa trên các tập dữ liệu của nhà máy đích.

3.3.3. Giải thích đóng góp của đặc trưng bằng LOFO và SHAP trong quá trình chuyển giao

Phân tích SHAP tiếp tục được sử dụng để khảo sát sự biến đổi của mẫu quy gán đặc trưng khi tăng FT. Trong chuyển giao zero-shot (zero-shot transfer) ($FT = 0$), EPSc là đặc trưng xếp hạng cao nhất. Biến này đóng góp tỷ trọng lớn nhất vào tổng mức quy gán (Fig. 6(b); Fig. S8(a)). Mức đóng góp này vượt xa Chất rắn lơ lửng bay hơi hỗn hợp bùn lợ (MLVSS) và EPSp. Điều này cho thấy trong chuyển giao zero-shot, mô hình được chuyển giao duy trì mẫu quy gán tập trung mạnh vào EPSc. Đặc điểm này phản ánh sự tái hiệu chỉnh hạn chế trước khi tinh chỉnh (fine-tuning). Khi FT tăng từ 10% đến 30%, mẫu quy gán trở nên phân tán hơn. Mức quy gán cho EPSp tăng lên, cùng với các đóng góp tương đương hơn từ EPSc, SRT và MLVSS (Fig. S9–S10). Tại $FT = 40\%$, EPSc và EPSp xếp hàng đầu và cùng chiếm hơn nửa tổng mức quy gán (Fig. 6(c); Fig. S8(b)). Kết quả này thể hiện một cấu trúc quy gán cân bằng hơn tập trung vào EPS sau khi tinh chỉnh. Thứ hạng đặc trưng và cấu trúc quy gán tại $FT = 50\%$ phần lớn tương tự như tại $FT = 40\%$ (Fig. S11). Điều này phù hợp với mức cải thiện hiệu suất bổ sung rất hạn chế khi FT vượt quá 40%. Cùng với kết quả LOFO, các phát hiện này chỉ ra rằng hiệu suất LSTM-FT chủ yếu nhờ các biến liên quan đến EPS. Trong đó, EPSc giữ vị trí dẫn đầu và EPSp trở nên nổi bật hơn sau khi tinh chỉnh.

3.3.4. Vì sao học chuyển giao (transfer learning) giúp dự đoán bám màng chính xác ở nhà máy đích (target plant)?

Kết quả làm sạch tại chỗ (CIP) hai bước củng cố thêm mối liên hệ giữa phản ứng EPS gắn với Fe và thành phần lớp bám bản giữ lại ở nhà máy đích. Trong giai đoạn làm sạch bằng axit xitric (CA), tổng Fe tăng mạnh từ 0.62 lên 43.02 mg/L (Fig. 7(e)). Kết quả này cho thấy lượng lớn thành phần chứa Fe đã được giải phóng khỏi lớp bám bản. Trong giai đoạn làm sạch bằng NaClO kế tiếp, Tổng cacbon hữu cơ (TOC) tăng từ 5.19 lên 43.22 mg/L (Fig. 7(f)). Đồng thời, tổng Fe cũng tăng từ 1.85 lên 11.16 mg/L. Điều này phản ánh sự giải phóng của chất hữu cơ có thể oxy hóa cùng các hợp phần mạng Fe. Tại nhà máy đích, Nhu cầu oxy hóa cùng các hợp phần mạng Fe. Tại nhà máy đích, Nhu cầu oxy hóa học hòa tan (SCOD) thấp hơn Nhu cầu oxy hóa học tổng số (TCOD) (93 so với 271 mg/L). Các kết quả này chứng minh rằng lớp bám bản chủ đạo không chỉ do riêng chất hữu cơ hòa tan tạo thành. Thay vào đó, lớp bám bản gắn kết chặt chẽ hơn với cấu trúc nền hữu cơ liên kết với bùn chứa các hợp phần liên quan đến Fe.

Fig. 7. Bằng chứng hóa lý củng cố cho mẫu quy gán lấy EPS làm trung tâm do LSTM-FT học được tại nhà máy đích. Các đặc tính của hỗn hợp bùn lợ gồm (a) Thời gian hút mao dẫn (CST) cùng CST riêng và (d) độ nhớt cùng độ nhớt riêng. Phổ Ma trận kích thích - phát xạ (EEM) của (b) EPS và (c) SMP. Nồng độ sắt tổng số trong các dung dịch làm sạch thu thập trong quy trình làm sạch tại chỗ (CIP) hai bước thể hiện ở (e). Nồng độ Tổng cacbon hữu cơ (TOC) trong dung dịch làm sạch thu thập trong quá trình rửa bằng natri hypoclorit thể hiện ở (f). Trước CIP, cụm màng được xả cặn và tráng rửa bằng nước lọc qua màng. Quy trình CIP hai bước bao gồm làm sạch bằng axit xitric (15 g/L, pH = 2.5), tiếp theo là natri hypoclorit (1 g/L, pH = 10). Các mẫu được thu thập ngay trước và sau mỗi bước làm sạch. Ký hiệu B-CA và A-CA tương ứng biểu thị các mẫu lấy trước và sau khi làm sạch bằng axit xitric. Ký hiệu B-NaClO và A-NaClO biểu thị các mẫu lấy trước và sau khi làm sạch bằng natri hypoclorit.

Phân tích trước đó chủ yếu tập trung vào kịch bản MBR xử lý nước thải sinh hoạt. Do đó, nghiên cứu tiến hành đánh giá bổ sung trên một hệ MBR quy mô thí điểm xử lý nước thải công nghiệp hóa dầu và được phẩm hỗn hợp. Trường hợp này cung cấp thử nghiệm sơ bộ cho quy trình học chuyển giao (transfer learning) đã thiết lập dưới một nền nước thải khác biệt hơn. So với nhà máy đích (target plant) xử lý nước thải sinh hoạt, MBR công nghiệp thể hiện một kịch bản chuyển giao khó khăn hơn. Hệ thống này có thành phần nước đầu vào phức tạp hơn, tải trọng hữu cơ cao hơn và động học sinh khối do các xáo trộn vận hành chi phối. Đường biến thiên áp suất xuyên màng (TMP) của MBR công nghiệp thể hiện một xu hướng dài hạn rõ rệt (Fig. 8(a)). Đường này cũng ghi nhận nhiều biến động đột ngột từ các sự kiện vận hành thực tế. Điều này phản ánh hành vi bám bản màng phức tạp trong các điều kiện kỹ thuật thực tế.

Thông tin bám bản vẫn được chia sẻ giữa các kịch bản. Protein trong EPS (EPSp) (256.8 ± 126.7 mg/L) có cùng bậc độ lớn với các nhà máy nguồn (source plant) (184–229 mg/L). Đồng thời, Polysaccharide trong SMP (SMPc) (6.9 ± 5.0 mg/L) cũng nằm trong phạm vi của nhà máy nguồn (5.2–11.5 mg/L). Sự tương đồng này cho thấy một phần thông tin bám bản học từ các nhà máy nguồn vẫn có thể chuyển giao sang MBR công nghiệp. Khả năng chuyển giao này duy trì tốt ngay cả dưới một nền nước thải khác biệt đáng kể.

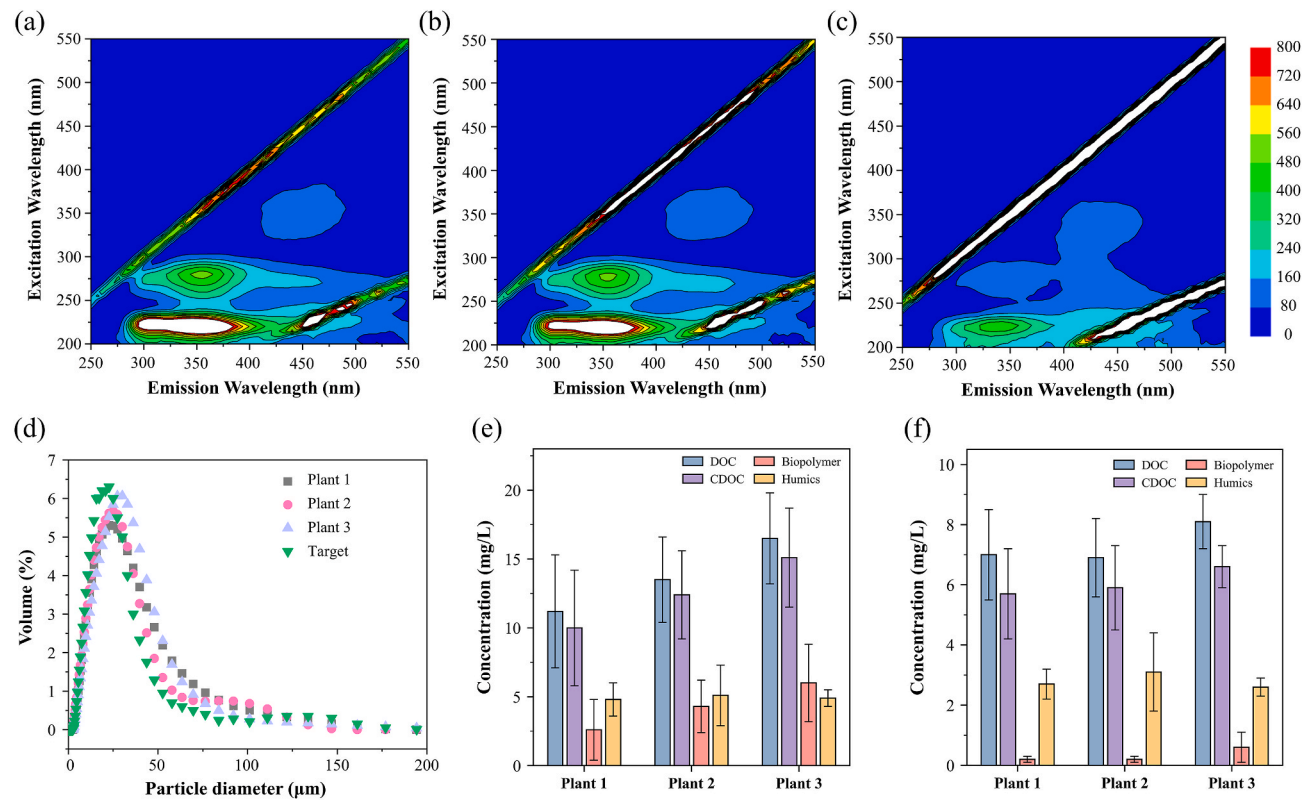


Fig. 4. Physicochemical evidence supporting the EPS- and SMP-related attribution pattern learned by LSTM-base in the source plants: EEM spectra of EPS from (a) Plant 1, (b) Plant 2, and (c) Plant 3; (d) sludge particle-size distributions of Plants 1–3 and the target plant; LC-OCD-derived organic fraction concentrations in (e) SMP and (f) effluent from Plants 1–3.

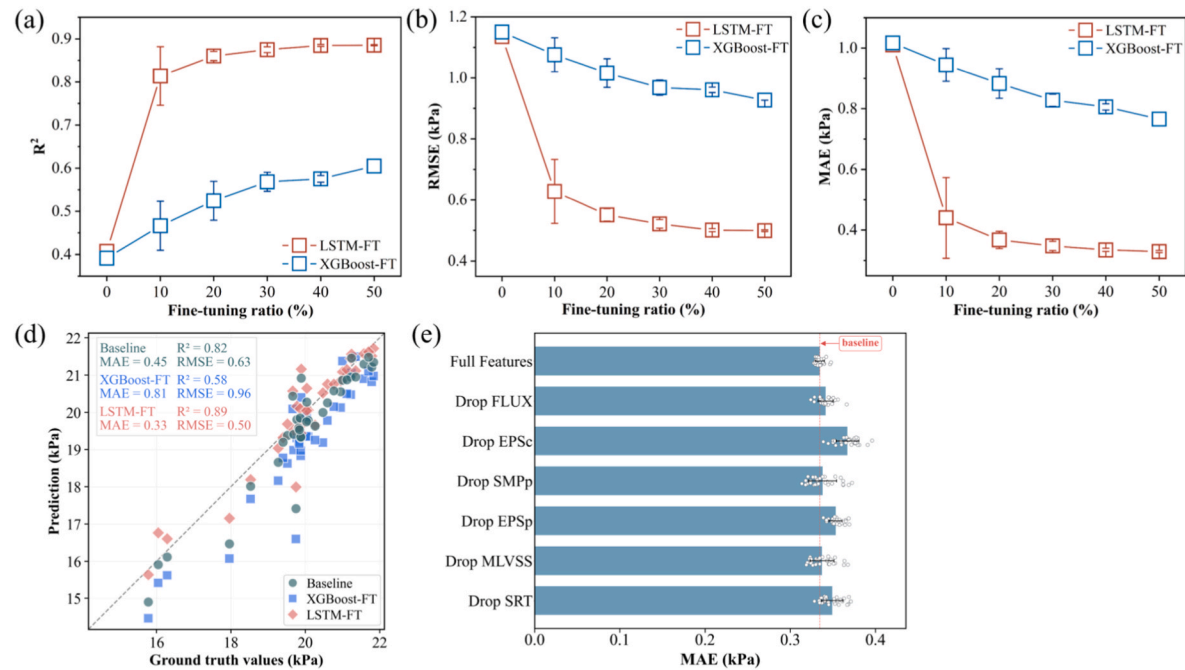


Fig. 5. Effect of fine-tuning ratio on target plant prediction performance and LOFO analysis: (a) R^2 , (b) RMSE, and (c) MAE as functions of fine-tuning ratio, (d) observed versus predicted TMP values for the baseline model, XGBoost-FT, and LSTM-FT on the target plant testing set, with the transfer models evaluated at FT = 40%, and (e) LOFO analysis results for LSTM-FT at FT = 40%. For each nonzero FT setting in (a)–(c), performance is reported as mean $\pm$ standard deviation across all fine-tuning segments and repeated runs, using up to six contiguous segments sampled from the fine-tuning pool and five random seeds for each retained segment.

FT = 40% was only marginal. In comparison, XGBoost-FT also improved with increasing FT, but the gains were smaller and more gradual, with

Fig. 8. Đánh giá bổ sung khung học chuyển giao (transfer learning) đề xuất trong MBR xử lý nước thải công nghiệp: (a) đường biến thiên áp suất xuyên màng (TMP) cùng các sự kiện vận hành chính, (b) thông kê tóm tắt các biến vận hành và hóa lý trong MBR công nghiệp, (c) mô hình LSTM tiền huấn luyện trước và sau khi tinh chỉnh, được đánh giá tại FT = 0 và FT = 40% bằng R^2 , MAE và RMSE, và (d) giá trị SHAP tuyệt đối trung bình của LSTM-FT tại FT = 40%.

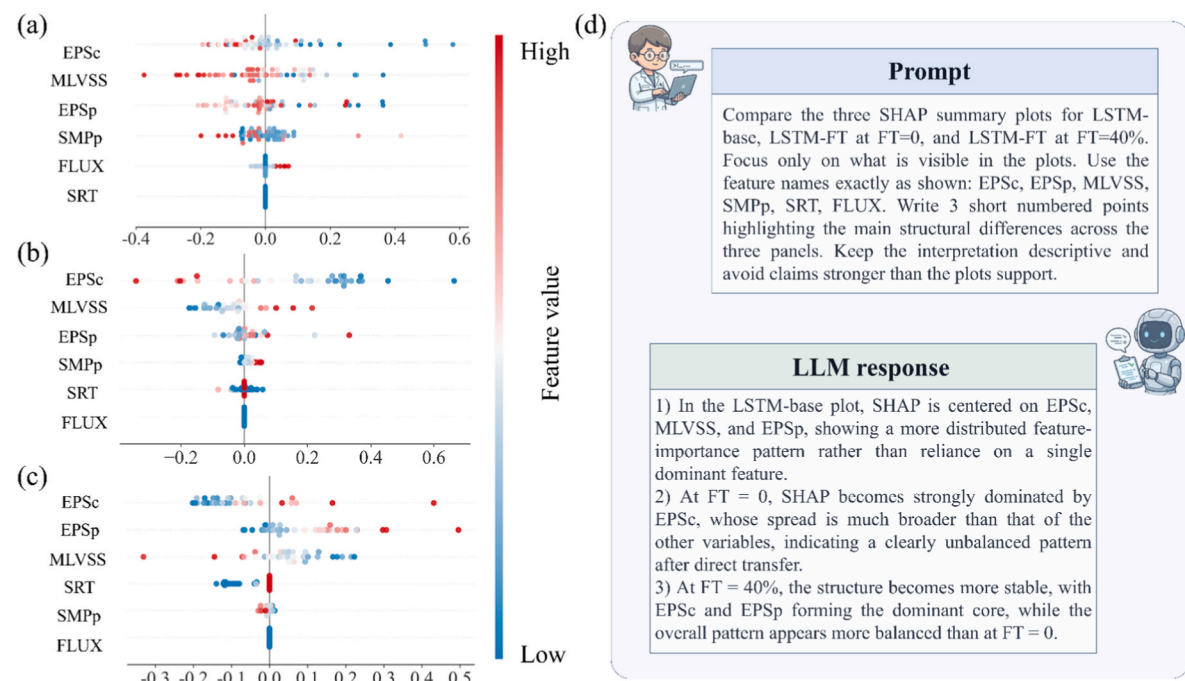


Fig. 6. SHAP attribution patterns and LLM-assisted summary for LSTM-base and LSTM-FT: SHAP summary plots of (a) LSTM-base, (b) LSTM-FT at FT = 0, (c) LSTM-FT at FT = 40%, (d) LLM-assisted summary of the corresponding SHAP observations.

R^2 increasing from 0.39 at FT = 0 to 0.58 at FT = 40% and only slightly further to 0.61 at FT = 50%. Overall, LSTM-FT showed larger performance gains at low FT levels than XGBoost-FT, and increasing FT beyond 40% provided limited additional improvement. Because both transfer models were adapted from models pretrained on the source plant datasets, these performance improvements should be interpreted as target plant adaptation through transfer learning under realistic data-limited conditions, rather than as conventional model development based solely on the 71 target plant records. In addition, for both transfer models the standard deviation across the fine-tuning segments and repeated runs was largest at FT = 10% and decreased at higher FT, becoming small by FT = 40%. This indicates that too few target records reduced the robustness of adaptation, whereas a moderate amount of target data was sufficient for stable performance. Accordingly, FT = 40% was selected as the representative fine-tuning condition for the subsequent analyses.

3.3.2. Performance comparison of baseline model and transfer learning models

To compare predictive performance on the target plant, the baseline model (specifically trained based on target plant datasets) together with LSTM-FT and XGBoost-FT was evaluated (Fig. 5(d)).

The testing samples were mainly distributed within a TMP range of approximately 19–21.5 kPa, with fewer samples at lower TMP values of around 15.5–18 kPa. Within the main TMP range, LSTM-FT showed the tightest clustering around the line of equality, indicating the best agreement between predicted and observed TMP values. The baseline model showed moderate deviations, whereas XGBoost-FT exhibited the largest scatter and the most evident departure from the line of equality. These patterns suggest that transfer learning improved target plant prediction when implemented with LSTM, whereas XGBoost-FT did not provide a performance advantage under the present setting.

The quantitative metrics showed the same trend. Relative to the baseline model, LSTM-FT reduced RMSE from 0.63 to 0.50 kPa and MAE from 0.45 to 0.33 kPa, corresponding to decreases of 20.6% and 26.7%, respectively, while increasing R^2 from 0.82 to 0.89. By contrast, XGBoost-FT increased RMSE to 0.96 kPa and MAE to 0.81 kPa, and

reduced R^2 to 0.58. Taken together, these results indicate that LSTM-FT achieved the best predictive performance under limited target plant data, outperforming both the baseline model and XGBoost-FT. Therefore, the following sections focus on LSTM-FT to further interpret the observed transfer behavior using LOFO and SHAP analyses.

3.3.3. LOFO and SHAP interpretation of feature contributions during transfer

LOFO and SHAP analyses were used to identify the variables that supported transfer performance in LSTM-FT. LOFO analysis was first conducted to evaluate the effect of each input variable on LSTM-FT performance at FT = 40% (Fig. 5(e)). The reference model was the LSTM-FT model trained with all input variables, which achieved an MAE of 0.33 kPa. Each variable was then removed individually, and the resulting MAE was compared with that of the reference model using paired two-sided Wilcoxon signed-rank tests (Table S10). Removing EPSc caused the largest performance loss, increasing MAE to 0.37 kPa ($p \leq 0.001$), indicating that EPSc was the most important variable for maintaining LSTM-FT performance. Removing EPSp or SRT also increased MAE to 0.35 kPa (both $p \leq 0.001$), whereas removing FLUX caused a smaller but still significant increase ($p \leq 0.01$). Overall, the LOFO results confirmed that EPS-related variables, especially EPSc and EPSp, were important for LSTM-FT performance, while SRT and FLUX also showed smaller but statistically significant effects.

SHAP analysis was further used to examine how the feature-attribution pattern evolved with increasing FT. Under zero-shot transfer (FT = 0), EPSc was the top-ranked feature, contributing the largest share of the total attribution, well above MLVSS and EPSp (Fig. 6(b); Fig. S8(a)). This suggests that, under zero-shot transfer, the transferred model retained a strongly EPSc-centered attribution pattern, reflecting limited recalibration before fine-tuning. From FT = 10% to 30%, the attribution pattern became more distributed, with increased attribution to EPSp and more comparable contributions from EPSc, SRT, and MLVSS (Fig. S9–S10). At FT = 40%, EPSc and EPSp were the two top-ranked features and together accounted for more than half of the total attribution (Fig. 6(c); Fig. S8(b)), indicating a more balanced EPS-centered attribution structure after fine-tuning. The feature ranking and

Tuyên bố đóng góp của tác giả theo CRediT

4. Ý nghĩa và triển vọng

• Việc xác thực trên MBR công nghiệp chứng minh thêm rằng khung làm việc vẫn hiệu quả dưới một bối cảnh bám bản khác biệt rõ rệt. Khác với nhà máy đích xử lý nước thải sinh hoạt, SMPp trở thành đặc trưng quan trọng thứ hai trong MBR công nghiệp. Điều này phản ánh tín hiệu SMP chứa protein nổi bật hơn, gắn liền với các xáo trộn vận hành và sự phân rã sinh khối. Vai trò vượt trội liên tục của EPSc trong MBR công nghiệp củng cố thêm tính ổn định xuyên kịch bản của nền tảng bám bản có thể chuyển giao này. Đồng thời, sự gia tăng quy kết của SMPp cho thấy tình hình có thể tái điều chỉnh trọng số thích ứng. Nhờ đó, mô hình phù hợp với các yếu tố gây bám bản đặc thù của từng nhà máy.

• Nhìn chung, nghiên cứu này chứng minh rằng thông tin liên quan đến bám bản dùng chung có thể được chuyển giao giữa các nhà máy MBR. Thông tin này được tái hiệu chuẩn có chọn lọc với dữ liệu hạn chế từ nhà máy đích. Đồng thời, các bằng chứng lý hóa độc lập giúp quá trình thích ứng này có thể giải thích được về mặt khoa học. Do đó, khung nghiên cứu cung cấp một nền tảng thực tiễn để dự đoán TMP đáng tin cậy. Khung này hỗ trợ ra quyết định kiểm soát bám bản thích ứng với từng nhà máy trong các hệ thống MBR hạn chế dữ liệu.

5. Kết luận

attribution structure at FT = 50% remained largely similar to those at FT = 40% (Fig. S11), consistent with the limited additional performance improvement beyond FT = 40%. Together with the LOFO results, these findings indicate that LSTM-FT performance was mainly supported by EPS-related variables, with EPS_c remaining top-ranked and EPS_p becoming more prominent after fine-tuning.

3.3.4. Why did transfer learning enable accurate membrane fouling prediction in the target plant?

Because SHAP and LOFO identify model-relevant variables rather than causal mechanisms, the attribution changes were interpreted against independent physicochemical evidence from the target plant. At FT = 0, the attribution pattern remained dominated by EPS_c, and the model retained moderate predictive performance ($R^2 = 0.41$), indicating that the EPS_c-centered predictive structure learned by the base model was partially applicable to the target plant. The relatively high EPS_c concentration in the target plant (49.6 mg/L) further suggests that polysaccharide-related fouling remained important. Elevated polysaccharide content can enhance water retention and reduce sludge dewaterability, thereby promoting the formation of a highly hydrated, compressible, and high-resistance fouling layer. This interpretation was consistent with the relatively high CST (49.45 s) and specific CST (4.36 s/(g/L)) observed in the target plant (Fig. 7(a)). Taken together, these results suggest that EPS_c represented a shared membrane-fouling matrix characteristic across plants that remained stably informative during transfer prediction.

Performance improved when FT increased to 40% ($R^2 = 0.89$), and the most notable change in the SHAP results was the rise of EPS_p to the second most important feature. This shift suggests that fine-tuning increased model sensitivity to protein-related EPS characteristics in the target plant. The EPS EEM spectrum of the target plant showed a dominant aromatic protein-like peak (Fig. 7(b)), consistent with the protein-related EPS signature observed in the source plants. Of note, it

was hypothesized that the increased importance of EPS_p (aromatic proteins) could be due to the Fe-enriched influent in MBRs [54]. To further support this interpretation, two Fe-dosing stages in the target plant were compared (Table S11). Specifically, upstream Fe was dosed into the influent mixing tank before pretreatment at target dosages of 9 and 26 mg/L, allowing Fe accumulation and EPS compositional responses in the MBR system to be examined. As the Fe dosage increased, the sludge-associated Fe content increased from 45 to 101 mg/g-MLVSS, while EPS_p increased from 47.96 to 62.30 mg/g-MLVSS. These observations suggest that Fe-enriched conditions were associated with a more protein-enriched EPS composition, which may explain the increased attribution of EPS_p after fine-tuning. Higher EPS protein content is generally associated with stronger hydrophobicity and greater floc aggregation, which may favor membrane-surface adhesion and cake-layer densification [46,55]. This interpretation was consistent with the relatively high viscosity and specific viscosity of the target sludge (50.03 mPa s and 9.46 mPa s/(g/L), respectively; Fig. 7(d)). Taken together, the increased SHAP attribution of EPS_p suggests that fine-tuning recalibrated the model toward the protein-related EPS characteristics that were more pronounced under the Fe-enriched conditions of the target plant.

The two-step CIP results further supported the association between the Fe-related EPS response and the composition of the retained fouling layer in the target plant. During the citric acid (CA) cleaning stage, total Fe increased sharply from 0.62 to 43.02 mg/L (Fig. 7(e)), indicating the release of substantial Fe-bearing components from the fouling layer. During the subsequent NaClO cleaning stage, TOC increased from 5.19 to 43.22 mg/L (Fig. 7(f)), while total Fe also increased from 1.85 to 11.16 mg/L, suggesting the release of oxidizable organic matter together with Fe-bearing components. Together with the lower SCOD than TCOD in the target plant (93 vs. 271 mg/L), these results suggest that the dominant fouling layer was not formed by dissolved organics alone, but was more closely associated with a sludge-bound organic matrix

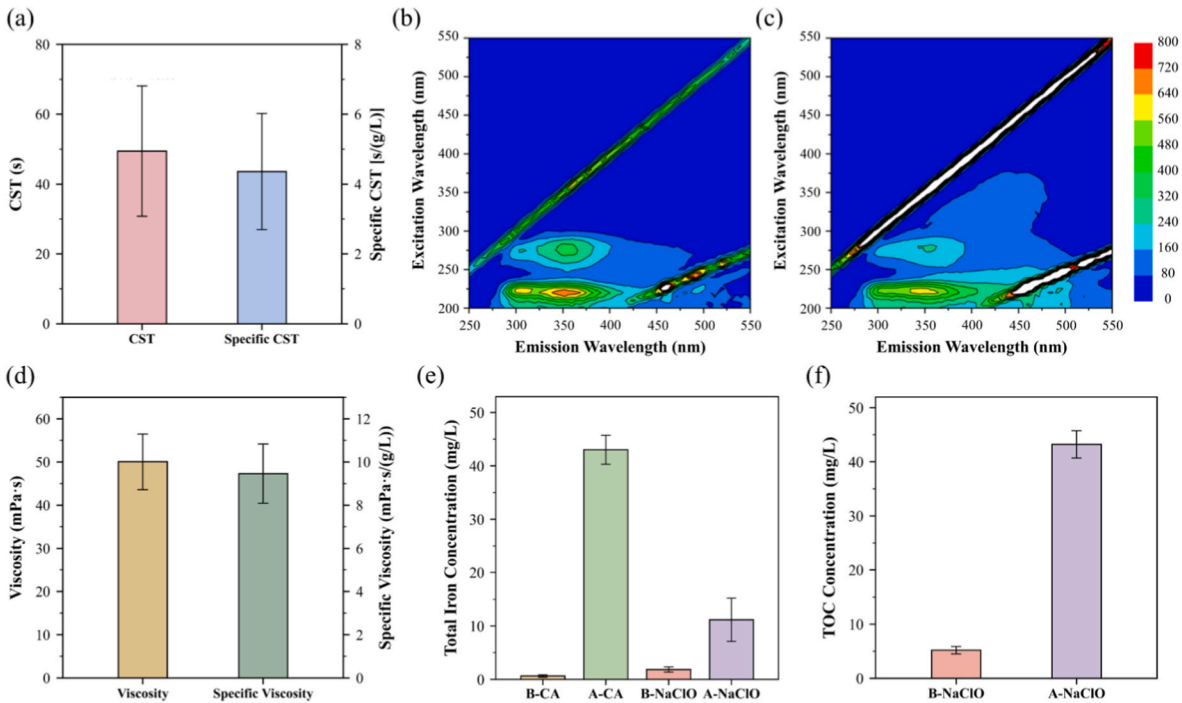


Fig. 7. Physicochemical evidence supporting the EPS-centered attribution pattern learned by LSTM-FT in the target plant: Mixed-liquor properties, including (a) CST and specific CST and (d) viscosity and specific viscosity; EEM spectra of (b) EPS and (c) SMP; (e) total iron concentrations in cleaning solutions collected during the two-step clean-in-place (CIP) procedure, and (f) TOC concentrations in cleaning solutions collected during sodium hypochlorite cleaning. Before CIP, the membrane unit was drained and rinsed with filtrate. The two-step CIP consisted of citric-acid cleaning (15 g/L, pH = 2.5) followed by sodium hypochlorite cleaning (1 g/L, pH = 10), with samples collected immediately before and after each cleaning step. B-CA and A-CA denote samples collected before and after citric-acid cleaning, respectively, whereas B-NaClO and A-NaClO denote samples collected before and after sodium hypochlorite cleaning.

Dữ liệu sẽ được cung cấp khi có yêu cầu hợp lý.

Tuyên bố về xung đột lợi ích

Phụ lục A. Dữ liệu bổ sung

Các tác giả tuyên bố không có xung đột lợi ích tài chính hoặc quan hệ cá nhân nào có thể ảnh hưởng đến công trình trong bài báo này.

Lời cảm ơn

Dữ liệu bổ sung cho bài viết này có thể được tìm thấy trực tuyến tại <https://doi.org/10.1016/j.memsci.2026.126065>.

Các tác giả trân trọng cảm ơn sự hỗ trợ tài chính từ Quỹ Học giả Thái Sơn của tỉnh Sơn Đông theo Mã số Tài trợ tsqn202312222. Nghiên cứu này cũng được hỗ trợ bởi Quỹ Khoa học Tự nhiên Quốc gia Trung Quốc (42406133). Ngoài ra, Quỹ Nghiên cứu Cơ bản và Nghiên cứu Cơ bản Ứng dụng tỉnh Quảng Đông (2023A1515110786) cũng tài trợ cho nghiên cứu. Các tác giả cũng cảm ơn các thành viên nhóm nghiên cứu của Giáo sư How Yong Ng tại Singapore (như Wei Hao Loh, David Imanuel Tanaka, v.v.). Nhóm đã hỗ trợ quý báu trong việc thu thập dữ liệu vận hành từ các nhà máy bể phản ứng sinh học màng (MBR) quy mô thử nghiệm. Dữ liệu được sử dụng riêng cho việc mô hình hóa và mô phỏng dựa trên AI trong nghiên cứu này. Mọi thông tin nhạy cảm, bảo mật hoặc mang tính nhận dạng đều được ẩn danh hoặc che giấu trước khi phân tích.

Tính sẵn có của dữ liệu

containing Fe-related constituents.

Furthermore, the decline in SMPp attribution after fine-tuning suggests that LSTM-FT placed less emphasis on soluble protein-related information during target plant adaptation. Although the SMP EEM spectrum of the target plant showed a pronounced low-excitation aromatic protein-like feature (Fig. 7(c)), the effluent EEM exhibited broader fluorescence regions ($E_x/E_m \approx 220\text{--}250/300\text{--}460\text{ nm}$) that partially overlapped with the low-excitation protein-like region in SMP (Fig. S12). This overlap suggests that part of the low-excitation protein-like dissolved organic signal was not fully retained by the membrane system. Therefore, the reduced attribution of SMPp after fine-tuning does not exclude its contribution to fouling, but suggests that SMPp was less closely associated with the dominant retained foulant layer in the target plant than EPS-related variables.

Taken together, LSTM fine-tuning preserved the transferable EPS-related fouling information in the target MBR plant, recalibrated the target-specific EPSp-related signal, and placed less emphasis on SMPp, which was less representative of the dominant fouling layer.

3.3.5. Cross-scenario adaptability to an MBR treating industrial wastewater

Because the preceding analysis focused mainly on the domestic MBR scenario, additional validation was conducted using a pilot-scale MBR treating mixed petrochemical and pharmaceutical industrial wastewater. This case provided a preliminary test of the established transfer learning workflow under a more distinct wastewater background. Compared with the domestic target plant, the industrial MBR presented a more challenging transfer scenario, with a more complex influent composition, higher organic loading, and disturbance-driven biomass dynamics. The TMP profile of the industrial MBR showed a clear long-term trend together with multiple abrupt perturbations associated with real operational events (Fig. 8(a)), reflecting complex membrane-fouling behavior under practical engineering conditions.

The industrial MBR differed substantially from the source plants in wastewater and sludge properties, with consistently higher TCOD, SCOD, MLSS, and MLVSS (Fig. 8(b)). Importantly, despite these background differences, EPSc ($43.6 \pm 20.2\text{ mg/L}$) remained close to the source plant range ($32.3\text{--}43.5\text{ mg/L}$), suggesting that EPSc-related

fouling information was still shared across scenarios. EPSp ($256.8 \pm 126.7\text{ mg/L}$) was also of the same order of magnitude as in the source plants ($184\text{--}229\text{ mg/L}$), and SMPc ($6.9 \pm 5.0\text{ mg/L}$) fell within the source plant range ($5.2\text{--}11.5\text{ mg/L}$). These overlaps suggest that part of the fouling-related information learned from the source plants could still be transferred to the industrial MBR, even under a substantially different wastewater background.

Under zero-shot transfer ($FT = 0$), the base model achieved only moderate performance in the industrial MBR ($R^2 = 0.46$; Fig. 8(c)), suggesting that the fouling-related information learned from the source plants was partially applicable but insufficient to fully capture the TMP dynamics in the industrial MBR. Fine-tuning with industrial MBR data improved prediction performance. This validation used the same chronological fine-tuning/testing logic, pretrained model, preprocessing procedure, and model settings as the main target plant analysis. At $FT = 40\%$, R^2 increased to 0.89, while MAE and RMSE decreased by 68.6% and 54.7%, respectively, relative to $FT = 0$. This improvement indicates that the established workflow could be adapted to the industrial wastewater background without additional model optimization.

SHAP analysis further showed that EPSc retained the highest attribution after fine-tuning, whereas SMPp increased to the second-highest attribution level (Fig. 8(d)). This pattern differed from that observed in the domestic target MBR plant, where fine-tuning mainly increased the attribution of EPSp. Thus, the two target cases shared stable EPSc-related fouling information, but differed in the feature requiring target-specific recalibration: EPSp in the Fe-enriched domestic MBR and SMPp in the industrial MBR. The increased attribution of SMPp suggests that fine-tuning recalibrated the model toward a more prominent soluble proteinaceous signal under the industrial wastewater background.

The increased attribution of SMPp after fine-tuning was mechanistically plausible in the industrial MBR. SMPp was higher in the industrial MBR ($27.6 \pm 17.4\text{ mg/L}$) than in the source plants ($10.8\text{--}13.8\text{ mg/L}$), indicating a stronger background of dissolved proteinaceous organics. Previous characterization of this industrial MBR showed pronounced SMP protein peaks during operational disturbances, together with decreases in MLSS and MLVSS, consistent with biomass loss, cell decay, and proteinaceous SMP release into the mixed liquor [56]. Moreover,

Tài liệu tham khảo

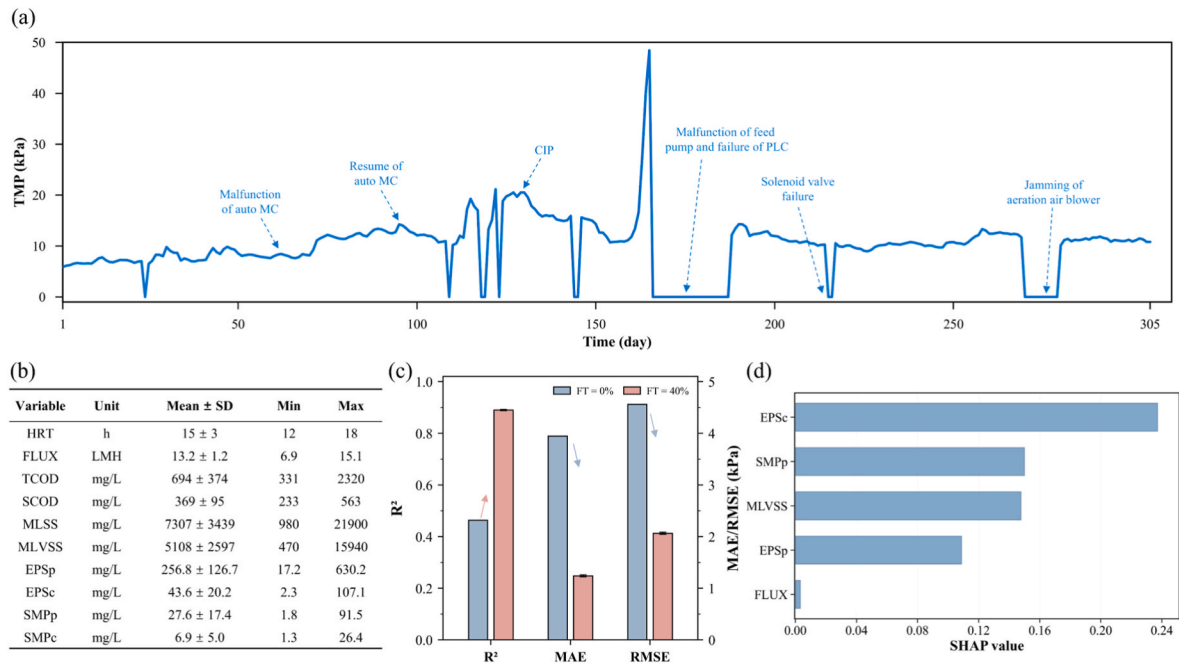


Fig. 8. Additional validation of the proposed transfer learning framework in an industrial wastewater MBR: (a) TMP profile with major operational events, (b) summary statistics of the operational and physicochemical variables in the industrial MBR, (c) the pretrained LSTM model before and after fine-tuning, evaluated at $FT = 0$ and $FT = 40\%$ using R^2 , MAE, and RMSE, and (d) mean absolute SHAP values of LSTM-FT at $FT = 40\%$.

SMP in this MBR was dominated by proteins rather than carbohydrates. These observations suggest that the enhanced SMPp attribution in the transfer model was associated with disturbance-driven soluble protein release. Such proteinaceous SMP could contribute to fouling development, thereby explaining the increased attribution of SMPp after fine-tuning.

Overall, this additional validation suggests that the proposed transfer learning workflow remained applicable to the industrial MBR, not because fouling mechanisms were identical across scenarios, but because the model retained transferable EPSc-related fouling information while fine-tuning recalibrated the attribution structure toward the dominant target-specific signal. In the domestic target MBR plant, this recalibration was reflected by the increased attribution of EPSp under Fe-enriched conditions, whereas in the industrial MBR, it was reflected by the increased attribution of SMPp associated with disturbance-driven soluble protein release. These results further support the ability of the transfer learning framework to preserve transferable fouling-related information while adapting to plant-specific fouling signals across distinct wastewater backgrounds.

4. Implications and outlook

The significance of this study lies not only in improving fouling prediction for a single pilot-scale MBR, but also in providing a basis for more mechanism-informed prediction in data-limited plants. By transferring knowledge learned from data-rich source plants, the LSTM model achieved reliable prediction in a data-limited target plant, suggesting that data-limited plants may not need to build fouling prediction systems entirely from scratch.

More importantly, the present results suggest that indicator selection for fouling prediction should begin with identifying the dominant fouling processes, followed by extraction of representative fouling fingerprints for prediction. Within this framework, indicators such as EPSc and EPSp appear to contain transferable fouling information across MBR plants. However, these variables are expensive, labor-intensive, and difficult to monitor routinely, which limits their direct practical use. A more feasible direction may therefore be to translate key fouling characteristics into monitorable surrogate fingerprints; in this regard, online spectral signals such as UV-vis and fluorescence fingerprints may provide a practical basis for plant-adaptive prediction and proactive fouling control [57].

Nevertheless, several limitations remain and point to directions for future work. The present models were developed from a limited number of pilot-scale MBRs with relatively similar operational backgrounds. At full scale, membrane fouling is further shaped by scale-dependent hydrodynamics and aeration and by greater operational variability than in pilot systems [58], so the pretrained relationships may not directly capture full-scale fouling dynamics. For systems with substantially different influent or operating conditions, the sludge and EPS/SMP composition and the dominant foulants can change, weakening the transferable EPS-centered signal and possibly requiring additional plant-specific indicators for reliable adaptation. The framework should therefore be further validated on larger datasets and extended to full-scale, geographically diverse plants to establish its robustness and generalizability. In addition, the prediction accuracy did not approach unity, indicating that the current indicator set is still incomplete. Besides organic foulants, the target plant also showed evidence of inorganic contributions, particularly iron-associated fouling revealed by the CIP results, which may not yet be fully captured by the present model inputs. Identifying more comprehensive indicators that better capture the combined effects of organic and inorganic foulants may in turn help support earlier fouling warning, better timing of operational adjustment and membrane cleaning, reduced energy and chemical consumption, and improved process stability (Fig. S13).

5. Conclusions

- Reliable base models for MBR fouling prediction were developed by pretraining on the three data-rich source plants. SHAP analysis showed that LSTM-base learned physically meaningful fouling-related patterns mainly through EPSc, EPSp, and SMPp, rather than relying on a single dominant operational signal. This was supported by the shared source plant characteristics, including EPS aromatic protein-like fluorescence, flocculated sludge particle-size distributions, and preferential retention of SMP by the membrane. These results indicate that LSTM-base provided a mechanistically interpretable and reliable pretrained basis for subsequent cross-plant transfer.
- Fine-tuning improved TMP prediction for the target plant treating domestic wastewater, with LSTM-FT achieving $R^2 = 0.89$ at 40% FT. SHAP and LOFO analyses showed that EPSc remained dominant, while EPSp gained importance after fine-tuning. Under the Fe-enriched condition of the target MBR plant, the increased importance of EPSp was associated with a shift toward protein-enriched EPS characteristics. Together, these findings explain why fine-tuning improved prediction accuracy: it retained the transferable EPSc-centered fouling basis while recalibrating the model toward Fe-associated, protein-enriched EPS characteristics that were more closely associated with fouling in the target plant.
- Validation in the industrial MBR further demonstrated that the framework remained effective under a markedly different fouling background. Unlike the target plant treating domestic wastewater, SMPp became the second most important feature in the industrial MBR, reflecting a more prominent proteinaceous SMP signal linked to operational disturbances and biomass decay. The continued dominance of EPSc in the industrial MBR further supported the cross-scenario stability of this transferable fouling basis, while the increased attribution of SMPp showed that fine-tuning could adaptively reweight plant-specific fouling drivers.
- Overall, this study demonstrates that shared fouling-related information can be transferred across MBR plants and selectively recalibrated with limited target-plant data, while independent physicochemical evidence makes this adaptation scientifically interpretable. The framework therefore provides a practical foundation for reliable TMP prediction and plant-adaptive fouling decision support in data-limited MBRs.

CRedit authorship contribution statement

Xiaohang Han: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **Liu Yang:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **Huan Qin:** Data curation, Investigation. **Shujuan Huang:** Data curation, Investigation. **Han Zhang:** Data curation, Investigation. **Boyan Xu:** Conceptualization, Funding acquisition, Project administration, Supervision, Writing – review & editing. **How Yong Ng:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.memsci.2026.126065>.

Data availability

Data will be made available on request.

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